

1.

- :
- : BERT
- : ,
- : https://github.com/lensraster/latest_lab_3

2.

BERT :

- ;
 - ;
 - ;
 - .
- , , .

3.

- :
- : bert-base-uncased ();
 - : distilbert-base-uncased ();
 - : ag_news (: World, Sports, Business, Sci/Tech);
 - CLI- : train predict;
 - : pytest;
 - : , , .
- :
- src/nlp_bert_classifier/pipeline.py — , , , , ;
 - src/nlp_bert_classifier/cli.py — ;
 - tests/ — ;
 - README.md — ;
 - reports/run_report.md — .

4.

4.1

```
python3 -m venv .venv
source .venv/bin/activate
```

```

pip install --upgrade pip
pip install -r requirements.txt
pip install -e .

```

4.2

```

pytest -q
: 8 passed.

```

4.3 (BERT)

```

python -m nlp_bert_classifier.cli train \
  --train-samples 200 \
  --test-samples 80 \
  --epochs 1 \
  --batch-size 8
artifacts/bert-ag-news/.

```

4.4

```

python -m nlp_bert_classifier.cli predict \
  --model-dir artifacts/bert-ag-news/model \
  --text "Apple releases a new AI chip for data centers"
JSON
.

```

4.5 (DistilBERT)

```

python -m nlp_bert_classifier.cli train \
  --model-name distilbert-base-uncased \
  --output-dir artifacts/distilbert-ag-news \
  --train-samples 200 \
  --test-samples 80 \
  --epochs 1 \
  --batch-size 8
artifacts/distilbert-ag-news/.

```

5.

- 1 (bert-base-uncased)
 - eval_accuracy = 0.6000
 - eval_f1_weighted = 0.5798
 - eval_loss = 1.1470
- 2 (distilbert-base-uncased)
 - eval_accuracy = 0.2750
 - eval_f1_weighted = 0.1717

```

- eval_loss = 1.3238
(1)
bert-base-uncased.

```

6.

7.

: BERT,